



Net Benefit Calculation Algorithm V7.1

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Wednesday, October 3, 2018

Version 7 is a significant change to the algorithm. See change log at the end of this document for changes between V7 and the previous version 6.1.

Objective

The Net Benefit Reward (NBR) Software is what is used to reward NBR to Producers after Consumers consume their products and services. At the heart of the software is the NBR formula or algorithm.

The objective of the formula is

1. Determine whether a given action is a “beneficial human action”. Beneficial human action is defined as a human action that results in benefit quantities which outnumber harm quantities that same action creates, as measured by the algorithm
2. In support of 1 above, ensure the software accounts for all inputs needed to perform the calculation
3. Give people a logical, rational way to calculate Net Benefit Rewards for any human action
4. Inspire human beings to act in Net Beneficial ways.

Goals

The goal of *this* document is provide clarity and transparency to how the algorithm calculates NBR, the inputs it uses, and how those inputs are weighted. For more information about the algorithm’s real time function, consult relevant sections on the algorithm and the Copiosis Organization (formerly known as the Payer Organization) at www.copiosis.com.

This proposal significantly updates the NBR algorithm proposed in “Net Benefit Calculation Algorithm Proposal V7” to Version 7.1. See the change log at the end of this document which lists changes between V7 and version 7.1.

Detailed discussion

Like most algorithms, the Copiosis NBR algorithm is a process described in a series of mathematical steps. These steps are happening constantly during a given calculation interval. They are also happening at different levels of society: Individual, organization, neighborhood, village, town, city, metropolis, region, nation, international, global.

It operates this way so cultural, regional and other differences between human communities can be captured in each calculation.

The variables

The formula is composed of letters (called variables) and symbols indicating mathematical operations. Some letters represent data community members collect and conclusions communities make. Researchers, observers and other specialists provide the data and communities make cultural decisions which effect the algorithm.

Specifically, people, sensors and other automated systems such as computers and cameras observe interactions between Producers and Consumers and how those interactions effect the environment. When a producer gives a consumer a product or service, and the consumer uses it, a transaction has occurred. When the results of that transaction are recorded, potentially, Net Benefit is produced. Unlike other economies, in Copiosis, a transaction is not fully complete until the transaction's Net Benefit is determined and rewarded.

The algorithm uses all these data and the community's cultural decisions to calculate the NBR Producers get.

Fine tuning variables

Other letters in the formula are not *data* placeholders. These letters are placeholders for numbers used to amplify one portion of the algorithm over another. These numbers are

important. They allow the Copiosis Organization to fine tune its Net Benefit Rewards to any human action, anywhere on the planet consistent with local customs and traditions. We call these variables *fine tuning variables*.

The data placeholder variables

Before getting into the full algorithm, let's first look at each variable. After we look at each variable, we'll gradually put them together until we end up with the full algorithm. This way you can understand how each element functions individually and alongside the others.

The data placeholder variables, as we wrote above, are replaced by numbers measuring actual conditions in the real world (research data). By turning these real world conditions into numbers, we can "measure then reward" human behaviors that create real world conditions we as a society prefer. We can also *not* reward human behaviors that create real world conditions we as a society do not prefer.

You can tell the algorithm and its output represents *subjective values* Copiosis believes fosters a vibrant human civilization which continually improves. This improvement theoretically happens with the minimum amount of negative side effects. We believe most of humanity will agree values expressed in the algorithm are appropriate given Copiosis' intent. If you disagree with these values, we sure would like to hear from you. We do not believe our values are perfect and thus unchangeable.

The following discussion of each variable also presents the Copiosis Organization's values, principles and beliefs. Societies and civilizations are based on fundamental principles or beliefs. We believe we have identified universal principles which can create an ideal society wherein everyone gets what they want no matter what one's perspective is. The key element which differentiates Copiosis from other systems is, *no person has authority to decide what is best for another person*. In other words, *everyone on the planet is free to express their life in the way they desire, so long as that expression does not impinge on the freedom of expression of others*.

We are offering these statements in full transparency. Anyone reading this proposal will therefore understand the moral values imbedded in Copiosis. You will find this statement at the bottom of each variable below.

The Human Benefit Variable

The Variable: hb

Former symbol from V6.1: S_h

What it measures: Effect on human resiliency within the calculation interval

What is its numerical range: -1-1

What is the numerical range means: The range represents a percentage of benefit with -1 indicating the human action destroys human resiliency, and 1 indicating the human action nurtures and develops human resiliency in the maximum extent possible.

What it tells us: To what degree does the result of a human action make *humanity as a whole* more resilient or less resilient?

Why Copiosis believes this is important: Human resiliency is a measure most disaster and disaster relief planners believe is one of humanity's critical characteristics. Resiliency is humanity's or an individual's ability to rebound from difficulties such as disasters, stress or other events. Copiosis believes a human civilization that is resilient is one that will last. A resilient civilization is made up of resilient community members as well as methods, models, technology and tools that contribute to strong, vibrant and resilient civilizations. We believe every human would agree an increasingly resilient civilization is something to nurture. You can read many resources about human resiliency [here](#).

The Environmental Benefit Variable

The Variable: eb

Former symbol: S_e

What it measures: Effect on the environment within the calculation interval

What is its numerical range: -1-1

What Environmental Benefit Variable range means: The range represents a percentage of benefit with -1 indicating the human action destroys the environment and makes it less sustainable, and 1 indicating the human action nurtures and develops the environment's life-supporting capability to the maximum extent possible.

What it tells us: To what degree does the result of a human action make the environment more sustainable/life supporting or less sustainable/life supporting?

Why Copiosis Believes this is important: A healthy environment that includes species diversity, clean air and water, plenty of natural land space and abundant natural resources is critical to humanity and the other animals we share the planet with. A healthy environment also supports human resiliency. Copiosis does not believe a healthy, vibrant industry or a wealthy population must come at the expense of the environment. The Copiosis economic system makes it possible for the two to coexist.

The Consumer Benefit Variable

The Variable: cb

Former symbol: $P_o/P(O)$

What it measures: Our best, scientific estimation of benefits to the consumer within the calculation interval

What is its numerical range: -1-1

What Consumer Benefit Variable range means: The range represents a percentage of benefit with -1 indicating that consuming this product or service is maximally destructive to human well being, and 1 indicating the human action nurtures and supports a consumer's well being.

What it tells us: It tells us science's best measure of benefits consumers get from a product/service.

Why Copiosis Believes this is important: A future society should offer a wide range of goods and services which benefit human well beings *while at the same time* offering as few as possible products and services that destroys well being. We believe there are two ways to collect data on this. One is through science-based measures. The other is the subjective experience of the consumer, after consuming the product or service. This variable represents data of the former category.

The Consumer Satisfaction Variable

The Variable: cs

Former symbol: $P_s/P(S)$

What it measures: Satisfaction a consumer experiences within the calculation interval from a given product or service

What is its numerical range: -1 -1

What the Consumer Satisfaction Variable range means: The range represents a percentage of benefit with -1 indicating that a given consumer is maximally *dissatisfied* with a product or service, and 1 indicating the consumer is maximally satisfied.

What it tells us: What the consumer's opinion of a product/service is.

Why Copiosis Believes this is important: We believe "satisfaction" comes in two parts: the benefits to a consumer's well being and the benefits the consumer believes they get. Producers should be rewarded for both. This variable allows consumers to have direct input into the algorithm (and thus the amount of NBR a producer gets) by providing their opinion about the good or service they have consumed measured in satisfaction.

The Consumer Producer Ratio

The Variable: $\text{Log}_e (1 + \text{NumberOfBeneficiaries/Producers})$

Former symbol: c/p and si

What it measures: The balance between the number of producers making something and comparing it to the number of people consuming that thing within a given calculation interval.

What is its numerical range: It measures actual populations. It is therefore bounded by actual counts.

What the Consumer Producer Ratio range means: The range represents a ratio comparing the number of producers (labor) making something *to the number of consumers consuming that thing*. with 1 indicating the least labor unit being employed. There is no upper bound on the range.

What it tells us: Within a given calculation interval, how productively efficient the Copiosis economy is, measured by the number of producers consuming resources to satisfy a given number of consumers. It also gives a sense of the scale of an action's impact.

Why Copiosis Believes this is important: We believe it's most sustainable to consume as few resources as possible (including labor) to make something while at the same time providing that thing to as many consumers as possible. This variable helps measure whether the economy optimizing efficiency from the perspective of labor.

Resource Abundance Variable

The Variable: ra

Former symbol: A

What it measures: A resource's renewal/depletion rate in a given calculation interval

What is its numerical range: 0 to no upper bound

What the Resource Abundance Variable range means: 0 indicates the resource is depleted. .5, for example, would indicate resources being depleted by 50 percent during the calculation interval. 1 would mean the resource is being renewed at the same rate it is being consumed. 2 - the resource is being renewed at twice the consumption rate.

What it tells us: It is a measure of sustainability. What is the rate at which resources are being used, compared to how they are being replaced in the time the measure is being taken?

Why Copiosis Believes this is important: An efficient economy is composed of industries which use the fewest resources possible for the population served. It also finds way to replenish resources used, prioritizing renewable resources use over non-renewable resource use. This variable helps measure whether the economy is optimized from the perspective of resources used (not including labor).

The Four Fine Tuning Variables

As we wrote above, these variables are important. That's because they allow the Copiosis Organization flexibility when calculating NBR rewards. These four variables fine tune up or down all the benefit measures (hb, eb, cb, and cs) individually.

We know this may introduce opportunity to reward human actions that are not net beneficial. We will be testing the algorithm so we can discover such problems and address them. The algorithm is a living process.

In the meantime, the algorithm "rules" require that the sum of all fine tuning variables equal to one. This creates a relationship between the variables thus eliminating arbitrary values being assigned to these variables.

For example, if one variable is assigned a value of .5, and another .3 and a third .2, the fourth value must equal .0. ($.5 + .3 + .2 + 0 = 1$).

Fine tuning variables:

Chb - increases or decreases the human benefit variable's influence on the algorithm

Ceb - increases or decreases the environmental benefit variable's influence on the algorithm

Ccb - increases or decreases the consumer benefit variable's influence on the algorithm

Ccs - increases or decreases the consumer satisfaction variable's influence on the algorithm

Putting it all together

Now that we've looked at all the variables in the formula, let's look at how they go together. Three sections comprise the algorithm. They are:

1. Production Demand Balance

2. Consumer Benefit

3. Universal Benefit

The **Production Demand Balance** is composed of the Resource Abundance Variable (ra) and the Consumer Producer Ratio ($\log_e (1 + \text{NumberOfBeneficiaries}/\text{Producers})$). Both variables are multiplied together.

Production Demand Balance = Resource Abundance * $\log_e$ (Consumer Producer Ratio)

$$(ra \cdot \log_e(1 + \frac{\text{NumberOfBeneficiaries}}{\text{Producers}}))$$

Since multiplication is being used within this section, each variable amplifies, or diminishes the other depending on whether the variable's value is positive or negative. Product Demand Balance has a direct relationship with NBR.

So if the Product Demand Balance result is *positive* and increases, we expect to see an increase in the NBR received. If the Product Demand Balance result is *negative*, it, by itself, produces 0 NBR because the NBR calculation results in negative Net Benefit. This is appropriate because we want how something is produced and how many people are producing it to be one of the most important calculations. What resources are used and how sustainable they are is another important calculation. So the resource abundance variable also can cause the algorithm to produce zero NBR all by itself. Of course if an action benefits no one (0), that too can (and should) result in 0 NBR because it is a waste of resources and labor.

The **Consumer Benefit** section is composed of the Consumer Benefit Variable and the Consumer Objective Benefit Variable. These variables are each multiplied by their fine tuning variables then added together.

$$\text{Consumer Benefit} = Ccs * \text{Consumer Satisfaction} + Ccb * \text{Consumer Benefit}$$

$$(Ccs \cdot cs) + (Ccb \cdot cb)$$

Taken together, the two variables in the Consumer Benefit section, along with their fine tuning variables may be able to result in 0 NBR. If either consumer satisfaction or consumer benefit is low, or negative, each can cause a reduction in the amount of NBR awarded as well. But neither has as big an effect as the Production Demand section. If they are both positive and high, and the fine tuning variables are also high, all else being equal, they can have a significant effect, so long as the Production Demand section is also favorable, meaning, production is happening in the most efficient manner. As we wrote above, since they're added together, then multiplied by the Production Demand Balance result, they also can result in 0 NBR if their sum is negative all other factors being positive.

The **Universal Benefit** Section is composed of the Environmental Benefit Variable and the Human Benefit Variable and their fine tuning variables. Both are multiplied by their fine tuning variables, then added together.

$$\text{Universal Benefit} = Ceb * \text{Environmental Benefit} + Chb * \text{Human Benefit}$$

$$(Ceb \cdot eb + Chb \cdot hb)$$

The Universal Benefit section has the same effect on how much NBR is rewarded as the previous section. Since the two variables are added together, then multiplied by the Production Demand Balance result, they also can result in 0 NBR if their sum is negative all other variables are positive.

We have now identified all the elements in the Copiosis algorithm. To calculate Net Benefit Reward, we add together the Universal Benefit and Consumer Benefit sections then multiply that by the Production Demand Balance section.

$$\text{Net Benefit Reward} = \text{Production Demand Balance} * (\text{Consumer Benefit} + \text{Universal Benefit})$$

Now we can replace names for the algorithm sections with their variables. We arrive at the actual NBR algorithm:

$$NBR = (ra \cdot \log_e(1 + \frac{NumberOfBeneficiaries}{Producers})) \cdot [(Ccs \cdot cs) + (Ccb \cdot cb) + (Ceb \cdot eb + Chb \cdot hb)]$$

The diagram illustrates the components of the NBR formula. Below the formula, three colored boxes with callout lines point to specific parts of the equation:

- A green box labeled "Production Demand Balance" points to the term $(ra \cdot \log_e(1 + \frac{NumberOfBeneficiaries}{Producers}))$.
- A blue box labeled "Consumer Benefit" points to the term $(Ccb \cdot cb)$.
- An orange box labeled "Universal Benefit" points to the term $(Ceb \cdot eb + Chb \cdot hb)$.

On the next page we offer a table outlining each variable. The fine tuning variables are not included in this table.

Variable	Name	Tells us	Scale
ra	Resource abundance	It is a measure of sustainability. What is the rate at which resources are being used, compared to how they are being replaced in the time the measure is being taken?	0 to no upper bound
c/p	Consumer Producer Ratio	Within a given calculation interval, how productively efficient the Copiosis economy is, measured by the number of producers consuming resources to satisfy a given number of consumers.	1 to no upper bound
$\text{Log}_e(1 + \text{NumberOfBeneficiaries/Producers})$	Social Impact	What portion of the human population benefits from the action. Measured as $\text{log}_e(1 + \text{NumberOfBeneficiaries/Producers})$	NA
cs	Consumer satisfaction measure	What the consumer's subjective opinion of a product/service is.	-1-1
co	Consumer objective benefit (scientific)	It tells us science's best measure of objective benefits consumers get from a product/service.	-1-1
eb	Environmental impact (Benefit)	To what degree does the result of a human action make the environment more sustainable/life supporting or less sustainable/life supporting?	-1-1
hb	Human impact (benefit)	To what degree does the result of a human action make humanity as a whole more resilient or less resilient?	-1-1

Questions:

1. What are and where do the scales for each variable reside and who determines where on the scale a given variable should draw from given a specific product or service?

The scales exist as part of the formula in the spreadsheet documentation. The spreadsheet is currently being used to examine test scenarios. These scenarios ensure formula results are consistent with our expectations. Once the software is complete, these scales will be incorporated into the algorithm and used as values replacing the various variables.

2. Is there a way to automate how the variables are populated?

Yes. All but one of the variables can be automated through software and connected to data sets for each variable. Data sets are created and updated continuously by the Copiosis Organization. Some variables are fixed for a time period to be determined by the Copiosis Organization and Grand Jury-type *citizen juries*. These juries are convened at the end of every rolling period in the same way jury duties are. The juries listen to expert testimony then make decisions on subjective variable inputs for the following rolling period. Jury members earn Net Benefit Reward for their service.

3. How do we make this as easy as possible for the consumer and producer?

A department of the Copiosis Organization will be formed to educate anyone who wants to know about the net benefit calculation. Handhelds which function as the interface between citizenry and the Copiosis system will also provide background and in-the-moment information to help people choose their actions based on net benefit calculation. However, one doesn't need to know how the formula functions to receive NBR.

4. How do we (or do we need to) create a baseline of all given products and services?

Yes we do. We will start with the community demonstration projects by creating a base set of products and services based on our knowledge of those communities. Project participants, over time, will grow these databases as new or newly-discovered products and services are realized. Many databases needed already exist in governments, nonprofit organizations and think tanks and private companies. So it's just a matter of redirecting their use after the transition and continuing to update the databases in the way they have.

5. Who decides and how are decisions made about the fine-tuning variables?

See question 2 above.

Change log

Wednesday, October 31, 2018

- Changed Consumer subjective benefit to Consumer satisfaction measure to better reflect the assumed causality of our measuring process.
- Added a new constant and variable to replace social impact Log e
- Changed Consumer objective benefit to consumer benefit
- Combined the social impact variable and the consumer producer ratio, expecting social impact and the consumer producer ration measures the same things and are therefore duplicitous

Modifications between v6.1 and v.7 as of Friday, October 19, 2018:

- Significantly changed the entire to address inconsistencies in variable names, to organize variable differently to account for variable importance and to make the algorithm more intuitive.
- The Resource Abundance Variable is now known as “ra”
- The Producer Population Variable P and the Consumer Demand Variable D have been incorporated into the new Consumer Producer Ratio c/p
- The Social Benefit Variable “S” was renamed Social Impact Variable and assigned a new variable “si”
- The Consumer Subjective Satisfaction Variable Ps has been renamed Consumer Subjective Benefit and reassigned the variable “cs”
- The Consumer Objective Benefit Variable Po has been assigned the variable “co”
- The Environmental Impact Variable “Se” has a new variable “eb” and has been renamed Environmental Benefit Variable
- The Human Impact Variable “Sh” has a new variable “hb”

- The Social Impact Variable has been moved into the “Production Demand Balance” (PDB) portion of the algorithm. This ties social impact to economic efficiency instead of having it a part of “Consumer Benefit”
- A_{Max} has been eliminated from the algorithm.
- Fine tuning elements for each *section* have been eliminated.
- The Environmental Benefit section of the algorithm is now called the Universal Benefit section
- All PDB Variables are now multiplied together instead of added together. This makes each variable more impactful to the whole algorithm.
- Additionally, the PDB section is now multiplied against the Consumer Benefit and Universal Benefit sections so PDB has more influence on the algorithm.
- All fine tuning elements are now coefficients instead of denominators. Making it easier to understand their influence on benefit variables.
- The M_1 fine tuning variable is now C_{cs}
- The M_2 fine tuning variable is now C_{co}
- The M_3 fine tuning variable is eliminated as the environmental impact variable is now in the PDB
- The M_4 fine tuning variable is now C_{eb}
- The M_5 fine tuning variable is now Chb
- Eliminates the E (effort) variable
- Changes operators in the C_{PDB} expression from subtraction to addition
- Changes operators in the Consumer Benefit section to addition